Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية تقنية الهندسة الكهربائية

Student Name: _____
Summer Semester: 21/22
Final Exam

Course Name: Electrical Circuits (2)
Instructors: Dr. Abdallah Alzyoud

Date: 6-9-2022
Exam Time: 110 minutes

Choose the correct answer.

****Consider the circuit shown in Figure 1, where the current source has $I_s = 2\angle 0°$ A_{rms} and answer questions 1-4.**

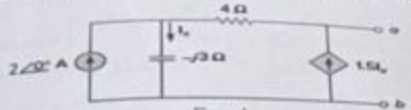


Figure 1

1. The open circuit voltage between the a-b terminals is :
   a. $-24+j12$ V b. $(25-j12)$ V c. $(12+j14)$ V d. None of all
2. The short circuit current (I_{sc}) when the a-b terminals are shorted is:
   a. -20 A b. $(-2.5+j2.6)$ A c. $(2.64+j0.48)A$ d. None of all
3. The Thevenin impedance (Z_{th}) between the a-b terminals is:
   a. $(6+j8)\ \Omega$ b. $(-8+j6)\Omega$ c. $(3+j2.5)\ \Omega$ d. None of all
4. The current through a load of $(6-j4)\ \Omega$ connected between the a – b terminals is:
   a. $(9+j3.)$ A b. $(8-j1)$ A c. $(15-j4)$ A d. None of all

****Consider the circuit shown in Figure 2 where $V = 10\angle -60°$ V_{rms}, and $I_s = 5\angle 10°$ A_{rms}, and answer questions 5-7.**

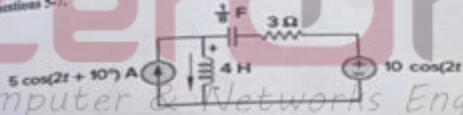


Figure 2

5. The voltage across the inductor due to *V* source only (in V) is:
   a. *12.9-j8.29* b. *14.71-j6.29* c. 12+j5 d. None of all
6. The voltage across the inductor due to *Is* source only (in V) is:
   a. -12.79-*j6.26* b. *40* c. *39.76-j4.36* d. None of all
7. The current through the inductor due to both sources in A is:
   a. *8.71+j6.29* b. *14.71-j6.29* c. 15.4−j3 d. None of all

*****Consider a 240 V rms, 60 Hz source supplies a load that is 10 kW(resistive),15kVAR(capacitive) and 22kVAR (inductive) and answer questions 8-12.**

8. The power factor of the load is
   a. 0.866 lagging b. *0.707 lagging* c. *0.82 lagging* d. None of all
9. The current of the load is
   a. *41.67 – j29.17 A* b. *34.75 – j12.0 A* c. *21 + j42.75 A* d. None of all
10. The apparent power supplied by the source is
    a. *52000 VA* b. 12210 *VA* c. *1231.5 + j564.8 VA* d. None of all
11. The capacitance required to improve the power factor to 0.96 lagging is
    a. *188.93 μF* b. *593.2 μF* c. *24.87 μF* d. None of all

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12. The current drawn from the source after the power factor correction is

a. 81.8 − j10.78 A b. 75 − j5.8A c. 41.67 − j12.15 A d. None of all

**Consider a balanced three phase Y- Δ system has $V_{an} = 120\angle 0°\ V_{rms}$ its positive sequence and a load of $Z_\Delta = 81 + j45\ \Omega$. If the line impedance is 0.4+j1.2Ω as shown in figure 3 and answer questions 13-17.



Figure 3

13. The voltage V_{an} is

a. 173 − j216.5 V b. 180 + j103.89 c. 240 V d. None of all

14. The line current I_a is

a. 5.69- j3.44 A b. 4.23+ j4.33 A c. − 2.63 − j3.3 A d. None of all

15. The phase current I_{ab} is

a. 3.54 + j0.54 b. 3.2+ j 73 A c. 2.84 − j0.63 A d. None of all

16. The total average power supplied by the source is

a. 4871 W b. 1329.8W c. 8437 W d. None of all

17. The total complex power consumed by the load is

a. 348 + j1.12 KVA b. 1.3 + j1.47 KVA c. 1.71 + j2.12 KVA d. None of all

**Consider a series RLC circuit has R = 4 Ω, L=25mH, C = 0.625pF, and $V_s = 15\angle 0°\ V_{rms}$ and answer questions 18-20 showing the required equations.

18. The resonant frequency (w_o) in (rad/sec) and the quality factor (Q) :.....

19. The cutoff frequencies w_1 , and w_2 (in rad/sec) are:.....

20. The voltage across the capacitor (V_C) at resonance (in V) is :.....

**Consider the magnetically coupled circuit shown in Figure 4 and answer questions 21-25 showing the required equations.

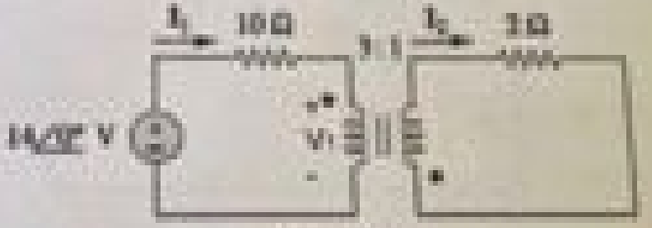


Figure 4

21. The equivalent input impedance (Z_{in}) seen by the source (in Ω):.....

22. The current supplied by the source (I_1) :.....

23. The voltage (V1) across the primary winding is:.....

24. The power absorbed by the 2Ω resistor is:.....

25. The complex power supplied by the source is:.....

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1. a 2. c 3. b 4. a 5. b

6. c 7. d 8. c 9. a 10. b

11. a 12. c 13. d 14. a 15. d

16. b 17. d

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18. $\omega_0 = \frac{1}{\sqrt{LC}} = 8000$, $Q = \frac{1}{R}\sqrt{\frac{L}{C}} = 50$

19. $Q > 10$ ✓ $\rightarrow$ $\omega_1 = \omega_0 - \frac{B}{2}$

$\omega_2 = \omega_0 + \frac{B}{2}$

$B = \frac{R}{L} = \frac{\omega_0}{Q} = 160\ H_z$

$\omega_1 = 7920$

$\omega_2 = 8080$

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Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

First Semester: 2021/2022
Final Exam

Course Name: Electrical Circuits (2)
Instructors: Dr A. Alzyoud and Eng J. Abu-Ghalune

Date: Jan. 31. 2022
Exam Time: 90 minutes

Prob., 1	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer													

Prob. 1: Choose the correct answer.

Consider the circuit shown in Figure 1, where the voltage source has $V_S = 5\angle 0°\ V_{rms}$.

1. The open circuit voltage (V_o) between the a-b terminals is
   a. $50(1-j1)$ V b. $75(1-j1)$ V c. $25(1-j1)$ V d. None of all
2. The short circuit current (I_{sc}) when the a-b terminals are shorted is
   a. -20 A b. -10 A c. -30 A d. None of all
3. The Thevenin impedance (Z_{th}) between the a-b terminals is
   a. $2.5 + j2.5\ \Omega$ b. $2.5 - j2.5\Omega$ c. $-2.5 + j2.5\ \Omega$ d. None of all
4. The current through a load of $7.5 - j2.5\ \Omega$ connected between the a – b terminals is
   a. $5(1-j1)$ A b. $10(1-j1)$ A c. $15(1-j1)$ A d. None of all

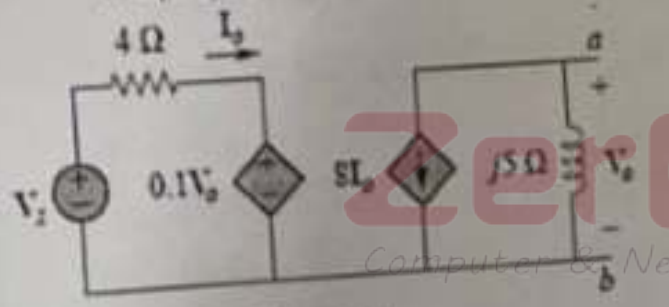


Figure 1

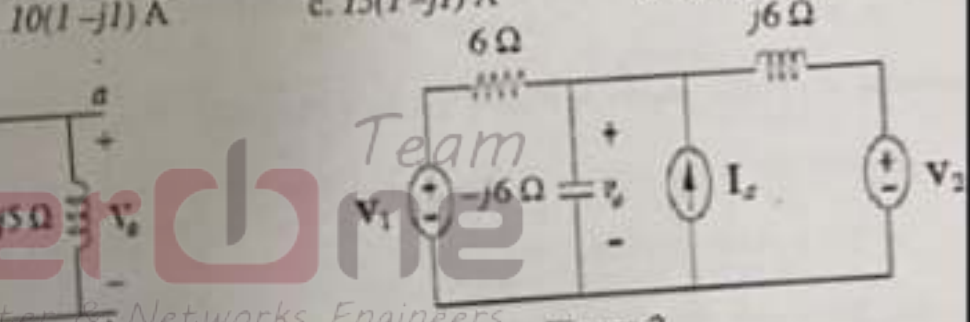


Figure 2

Consider the circuit shown in Figure 2 where $V_1 = 6\angle 0°\ V_{rms}$, $V_2 = 6\angle 0°\ V_{rms}$, ***and*** $I_S = 1\angle 0°\ A_{rms}$

5. The voltage across the capacitor due to V_1 source only (in V) is
   a. 6 b. 3 c. 12 d. None of all
6. The voltage across the capacitor due to V_2 source only (in V) is
   a. $-j12$ b. $-j6$ c. $-j3$ d. None of all
7. The voltage across the capacitor due to I_S source only (in V) is
   a. 6 b. 12 c. 18 d. None of all

Three Loads connected in parallel across a voltage source of $200\angle 0°\ V_{rms}$ ***where the loads have: Load 1: absorbs 250VA at 0.6 lagging p.f., Load 2: absorbs 200W at 0.8 leading p.f., and Load 3: ab 300VA and 150VAR.***

8. The power factor of the load 3 is
   a. 0.866 lagging b. 0.707 lagging c. 0.5 lagging d. None of all
9. The current of load 1 is
   a. $1.3 - j0.75$ A b. $0.75 - j1.0$ A c. $1 + j0.75$ A d. None of all
10. The complex power supplied by the source is
    a. $564.8 + j501.2$ VA b. $609.8 + j200$ VA c. $231.5 + j564.8$ VA d. None of all
11. The power factor of the equivalent load is
    a. 0.95 lagging b. 0.748 lagging c. 0.38 lagging d. None of all
12. The current supplied by the voltage source in rms is
    a. $2.822 - j2.506$ A b. $2.822 - j1.157$ A c. $3.049 - j1.0$ A d. None of all
13. The impedance of the equivalent load is
    a. $24.87 - j60.67\ \Omega$ b. $59.22 + j19.42\ \Omega$ c. $60.67 + j24.87\ \Omega$ d. None of all

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Prob.. 1	14	15	16	17	18	19	20	21	22	23	24	25	
Answer													

A balanced Δ-load of $Z_\Delta = 8.66 + j5\ \Omega$ is supplied by a balanced Y voltages of $V_{an} = 250/\underline{0^\circ}\ V_{rms}$ in positive sequence.

14. The voltage V_{AB} is
a. $375 + j216.5$ V b. $216.5 + j375$V c. 433 V d. None of all

15. The phase current I_{BC} is
a. $37.5 + j21.65$ A b. $j43.3$ A c. $-21.65 - j37.5$ A d. None of all

16. The line current I_a is
a. $37.5 + j65$A b. $j75$ A c. $37.5 - j65$ A d. None of all

17. The total average power supplied by the source is
a. 48.71 KW b. 28.21 KW c. 84.37 KW d. None of all

18. The total complex power consumed by the load is
a. $56.48 + j50.12$ KVA b. $56.48 + j23.15$ KVA c. $48.71 + j28.12$ KVA d. None of all

Consider a series RLC circuit has $R = 4\ \Omega$, $L = 4mH$, $C = 0.4\mu F$, and $V_s = 12/\underline{0^\circ}\ V_{rms}$

19. The resonant frequency (w_o) in (rad/sec) is
a. 25 K b. 50 K c. 20 K d. None of all

20. The quality factor (Q) is
a. 50 b. 30 c. 25 d. None of all

21. The cutoff frequencies w_1 and w_2 (in rad/sec) are
a. (19.2, 20.8) K b. (24, 26) K c. (49, 51) K d. None of all

22. The voltage across the capacitor (V_C) at resonance (in V) is
a. 400 b. 300 c. 12 d. None of all

Consider the circuit shown in Figure 3, where $V_S = 15/\underline{0^\circ}\ V_{rms}$, $R_1 = 2\ \Omega$, and $R_L = 4\ \Omega$.

23. The equivalent input impedance (Z_{in}) seen by the source (in Ω)
a. $4 + j3$ b. $3 + j3$ c. $5 + j3$ d. None of all

24. The current supplied by the source (I_1)
a. $2 - j1.2$ b. $1.6 - j1.2$ c. $2.5 - j2.5$ d. None of all

25. The power absorbed by the R_L resistor is
a. 12.5 W b. 10.88 W c. 8 W d. None of all

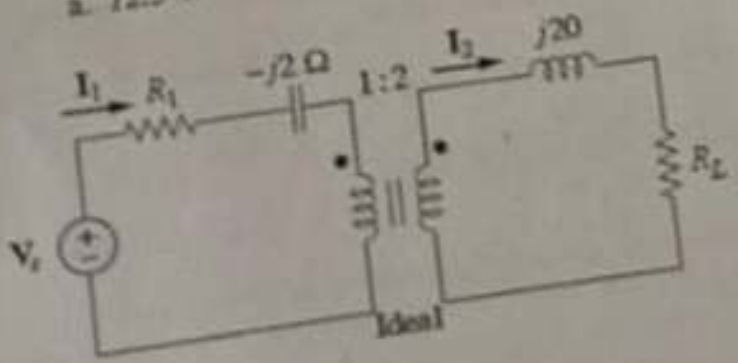


Figure 3

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$$I_o = \frac{V_s - 0.1V_o}{4} = \frac{5 - 0.1V_o}{4} \quad \cdots (1)$$

1-4

$$V_o = -8I_o \times j5$$

$$= -40jI_o$$

$$V_o = -40j\left(\frac{5-0.1V_o}{4}\right)$$

$$V_o = 25 - j25 \quad (c)$$

$$I_o = \frac{5 - 0.1V_o}{4}$$

$$I_{sc} = -8I_o = -8\left(\frac{5-0.1V_o}{4}\right) = -10\,A \quad (b)$$

$$Z_{th} = \frac{V_o}{I_{sc}} = -2.5 + j2.5\ \Omega \quad (c)$$

$$I = \frac{V_{th}}{Z_{th} + Z_L} = 5 - j5 \quad (a)$$

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$$V_{c1} = V_1 = 6\angle 0° \quad (a)$$

$$V_{c2} = V_2 \times \frac{3-j3}{3+j3}$$

5-7

$$= 6\angle -90 = -j6 \quad (b)$$

$$I_c = I_s \times \frac{3-j3}{3+j3} = 1\angle -90 = -j\,A$$

$$V_{c3} = I_c \times X_c = -j \times -6j = 6\,V \quad (a)$$

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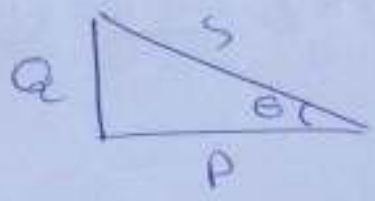
$V_{rms} = 200\angle 0°$

Load 1:-
$S = 250$ VA
0.6 lagging p.f

Load 2
$P = 200$ W
0.8 leading p.f

8-13

Load 3:-
$S = 300$ VA
$Q = 150$ VAR



$\sin\theta = \frac{Q}{S}$

$\theta = 30$ $\quad p.f = \cos(30) = 0.866$ (a)

I, ?? $\quad S = 250$



$\cos\theta = \frac{P}{S}$

~~$P = 250 \times 0.6 = 150$ W~~

$\tilde{S} = S\angle\theta = 250\angle 53.13°$

$\tilde{S} = \tilde{V}_{rms}\,\tilde{I}_{rms}^{*}$

$I_{rms} = \left(\frac{\tilde{S}}{\tilde{V}_{rms}}\right)^{*} = \frac{250\angle -53.13}{200\angle 0°} = 0.75 - j\,A$ (b)

$\tilde{S} = \tilde{S}_1 + \tilde{S}_2 + \tilde{S}_3 = 250 + 250 + 300\angle 18.16$

$= 641.8\angle 18.16°$ (b)

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$$\cos(18.16) = 0.95 \text{ lagging} \quad (a)$$

$$I_{rms} = \frac{S_{tot}^*}{V_{rms}} = 3.2091\angle -18.16^\circ \quad (c)$$

$$Z = \frac{V}{I} = 62.3247\angle 18.16 \quad (b)$$

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$Z_\Delta = 8.66 + j5\ \Omega$ P.P.S.

14-18

$V_{an} = 250 \angle 0^\circ V_{rms}$

$V_{ab} = V_{AB} = \sqrt{3}\, V_{an} \angle 30^\circ$

$= (\sqrt{3}\angle 30^\circ) \times (250\angle 0^\circ)$

$= 375 + j216.5$ v. (a)

I_{ac} ?

$I_{AB} = \frac{V_{AB}}{Z_\Delta} = \frac{375 + j216.5}{8.66 + j5} = 43 \angle -7.16^\circ$ A

$I_{CA} = I_{AB} \angle 120 = -21.65 + j37.5$ (a) ✓

$I_{AC} = -21.65 + j37.5$

$I_c = I_{cn} = I_{an} \angle 120$

$I_{an} = \frac{V_{an}}{Z_{\Delta/3}} \angle 120 = \frac{250\angle 120}{2.88 + j1.66} = j75$ (b)

$P = 3I^2 Z_\Delta = 3 \times 43.33^2 \times 8.66$

$= 48.71$ K watt.

$\tilde{S} = \frac{P}{\cos\theta} = 48.71 + 28.12j$ KVA (c)

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$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{9m \times 0.4\mu}} = 25000 = 25k$$ (a)

$$Q = \frac{\omega_0 L}{R} = \frac{25k \times 9 \times 10^{-3}}{4} = 25$$ (c) $Q \gg 10$ ✓

$$\omega_1 = \omega_0 - \frac{B}{2}, \quad \omega_2 = \omega_0 + \frac{B}{2}$$

19-22

$$B = 1000 = \frac{25000}{25} = \frac{\omega_0}{Q}$$

$$\omega_1 = 25k - 500 = 24.5k$$ (d)

$$\omega_2 = 25k + 500 = 25.5k$$

$$V_c = V_m Q = 12\angle 0^\circ \times 25 = 300$$ (b)

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$$Z_{in} = Z_R + R_1 + j\omega L_1 \qquad Z_R = \frac{Z_L}{n^2}$$

$$= 1 + j5 + 2 - j^2 \qquad = \frac{j20 + 4}{4} = 1 + j5$$

$$= 3 + j3$$ (b)

$$I_1 = \frac{V}{Z_{in}} = \frac{15\angle 0}{3 + j3} = 2.5 - j2.5$$ (c)

23-25

$$P = V_0 I_2$$

$$I_2 = \frac{I_1}{n} = +1.25 + j1.25$$

$$V_0 = I_2 \cdot R_L = +5 + j5$$

$$P = IV = 12.5\ W$$ (a)